

DAY 2 Milestone:

OVERVIEW :

☐ Dataset Generation & ML Model Development

Generate and pre-process synthetic historical data for rainfall, reservoir levels, and pressure.

Train and validate an ML model for water-level forecasting and risk classification.

Synthetic Dataset:

[smart_dam_dataset_updated.xlsx](#)

Dataset Description:

- Features:
 - rainfall
 - temperature
 - humidity
 - inflow
 - outflow
 - pressure
 - water_level
 - risk

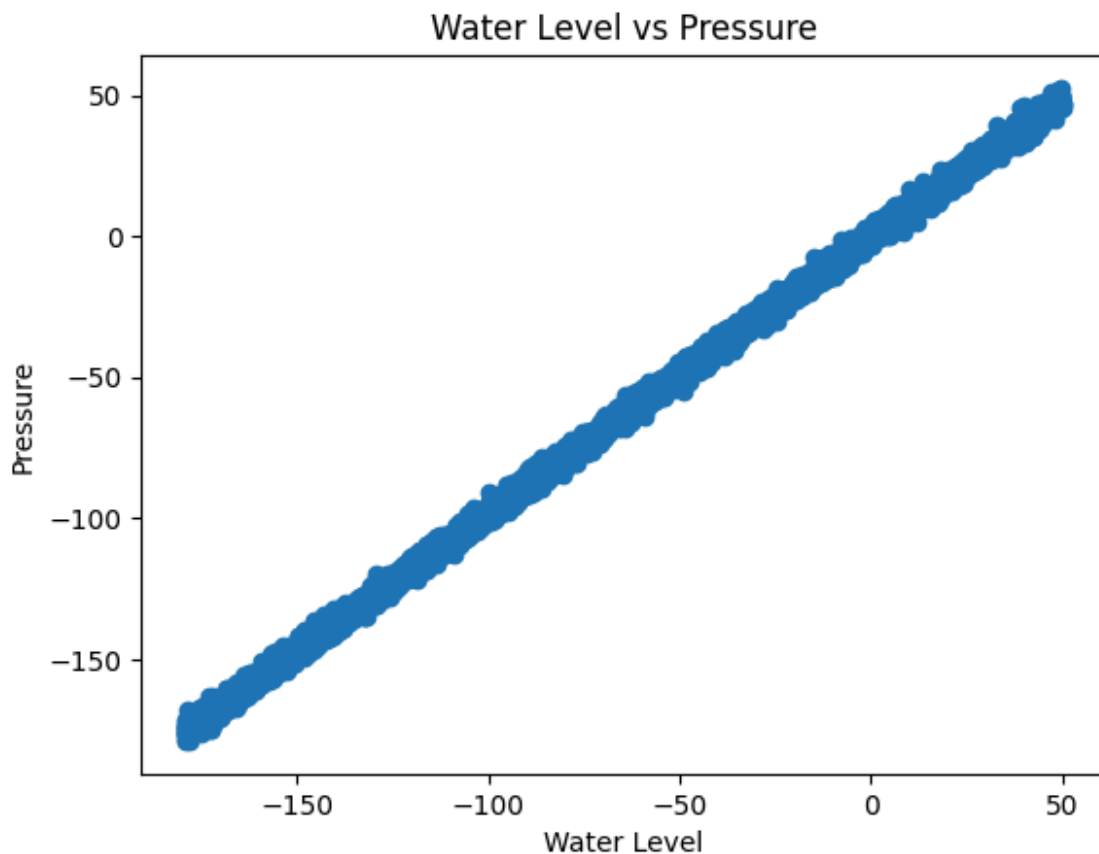
Machine Learning Model:

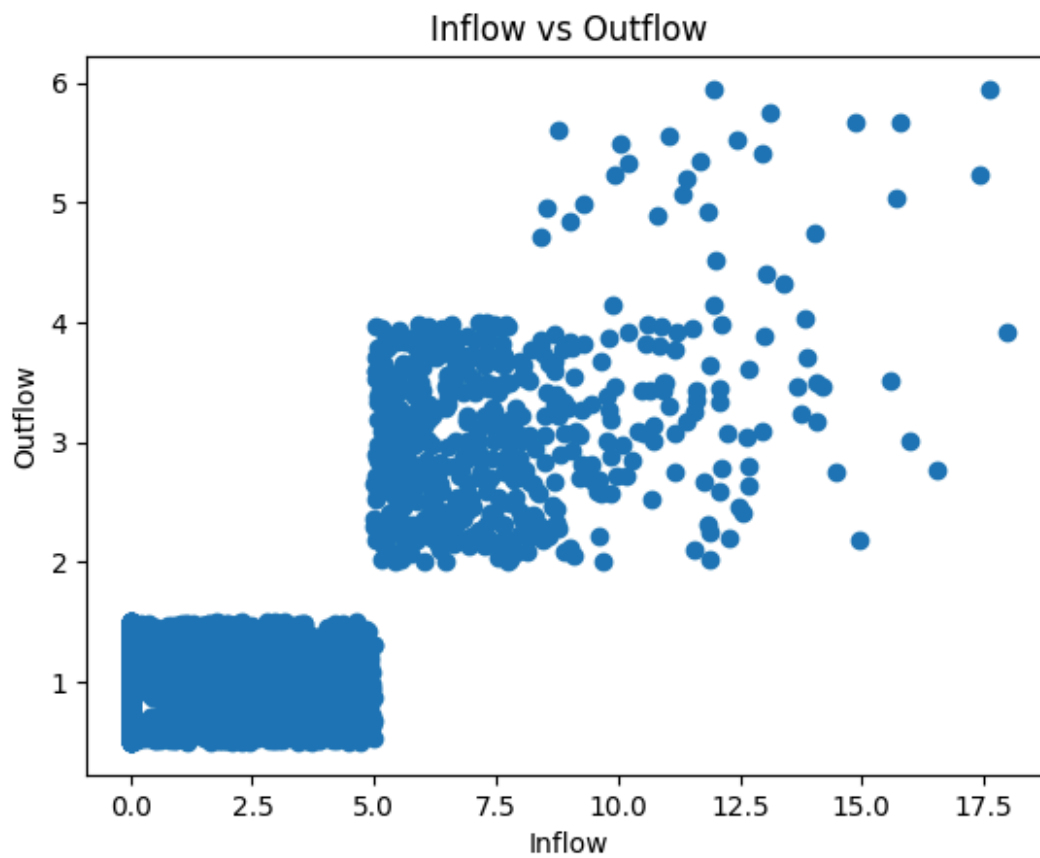
https://colab.research.google.com/drive/1Dzn0P6uV9gg8WqQk397ScB_-dRt3wOy7?sharingaction=ownershiptransfer

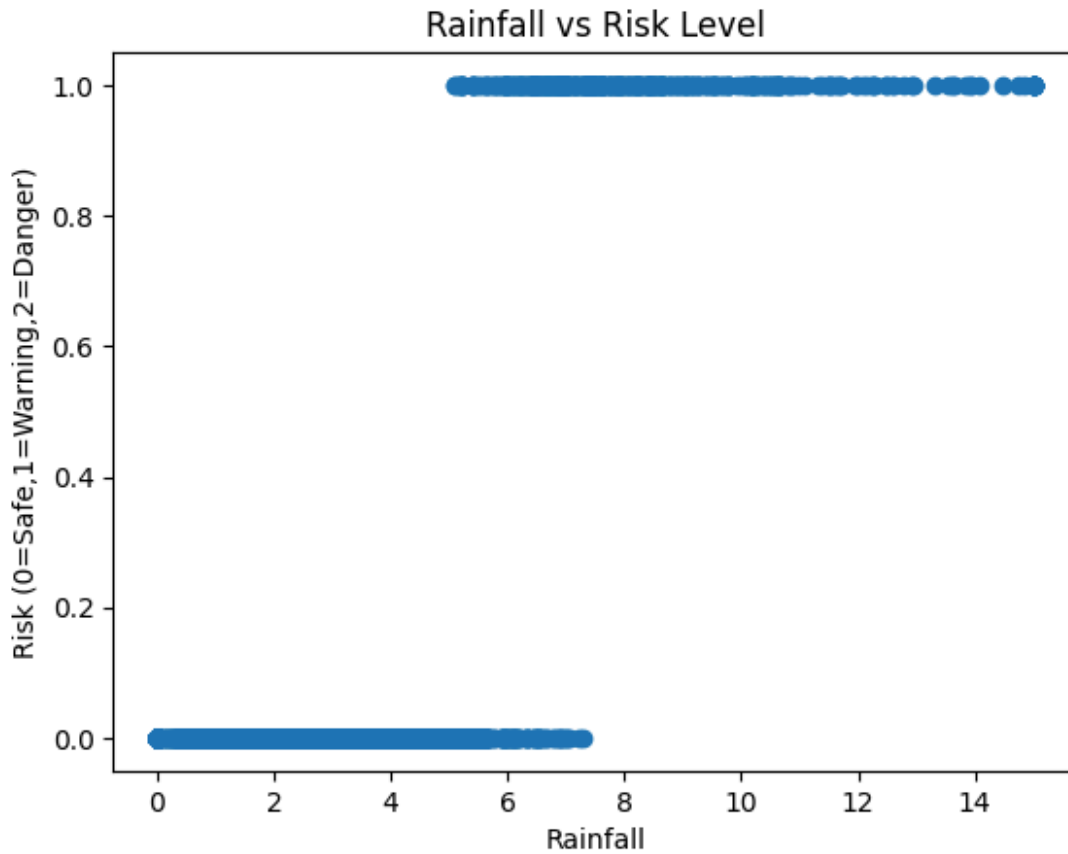
Exploratory Data Analysis (EDA):

Scatter plots:

- Rainfall vs Water Level
- Water Level vs Pressure
- Inflow vs Outflow







Model Implementation :

Multiple machine learning models were implemented to improve prediction accuracy and reliability.

1. LightGBM Model

- Used for regression (water level prediction)
- Handles non-linear relationships well
- Provides high accuracy

Use: Predict future water level

2. XGBoost Model

- Faster and efficient gradient boosting model
- Works well with large datasets

Use: Risk classification

3. LSTM Model (Time-Series)

- Deep learning model for time-series data
- Captures sequential patterns

Architecture:

- LSTM Layer (50 units)
- Dense Output Layer

Use: Forecast water levels over time

We combined outputs of multiple models:

Hybrid Model (Combined Approach)

- Initially used a simple average of predictions from:
 - XGBoost
 - LightGBM
 - LSTM
- After evaluation, optimized the hybrid model by selecting best-performing models
- Final Hybrid Model:
 - XGBoost → 60%
 - LightGBM → 40%
- Used weighted average to generate final prediction

Purpose:

- Predict **water level** in the dam system

- Classify **risk levels** based on predictions
- Improve accuracy using multiple ML models
- Models used:
 - XGBoost (tree-based)
 - LightGBM (tree-based)
 - LSTM (deep learning)

Model Comparison

- Evaluation metric used: **RMSE (Root Mean Squared Error)**

Model	RMSE	Performance
Hybrid (60% XGBoost + 40% LightGBM)	0.237	Best
LightGBM	0.316	Good
XGBoost	0.327	Good
LSTM	114.092	Poor (after rescaling)

Output:

	Model	RMSE
3	Hybrid	0.237150
1	LightGBM	0.316276
0	XGBoost	0.327356
2	LSTM	114.092032